

► 1 Task-based approaches to automation in the literature

Central to the study of technology on work is the insight that jobs are a “bundle of tasks” (Autor, 2015). As such, task automation might, or might not lead to job automation, depending on the importance of a particular task to an occupation. Analyses of employment effects of automation that use the task-based approach, including this study, attempt to capture two central considerations: (i) the ability of a specific technology to automate a given task and (ii) the character of existing occupations in the labour market, based on the detailed tasks performed by these occupations. By modelling the intersection of these two dimensions, one arrives at a range of possible projections of how such technology could interact with occupational tasks, leading either to their full automation through technology-driven substitution, or to a transformation, through partial automation and human-machine complementarity. While the task-based framework certainly has limitations, recent academic literature shows the analytical advantages of this method for modelling employment and labour market impacts (Acemoglu et al., 2024; Nurski and Ruer, 2024).

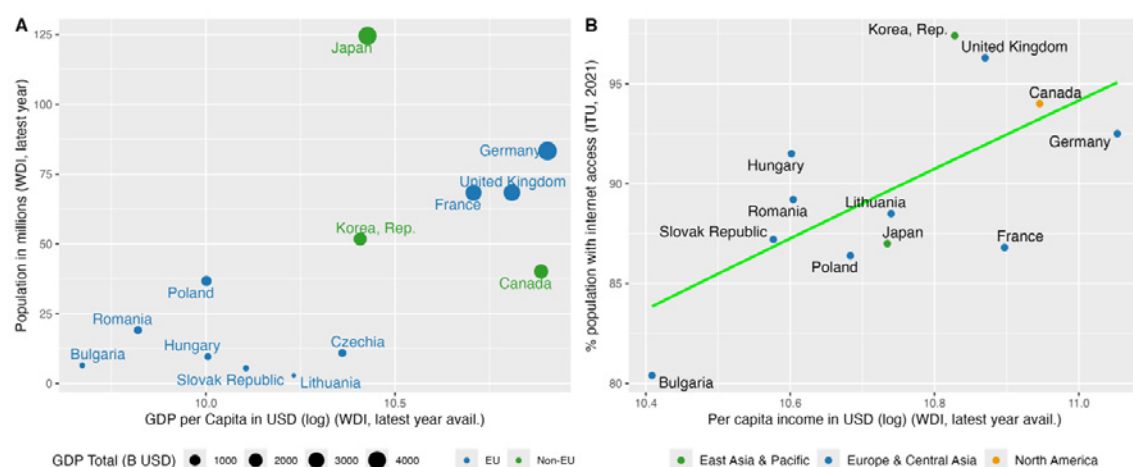
Research on AI and tasks has shown important improvements in methods and scope in recent years, which includes using AI tools for research purposes. In one of the early papers on Machine Learning (ML) and jobs, Frey and Osborne, (2013) fed the opinions of a group of IT technology experts into an ML algorithm to develop occupational-level projections of automation potential for the US job market. The emergence of GenAI APIs has simplified the possibility of combining human expertise and algorithmic capacities for research purposes. Relying on this idea, Eloundou et al. (2023) demonstrated a close alignment of GPT-4 predictions with a survey of 70 AI experts on the potential of automating occupational tasks with LLMs, and subsequently built on this feature to develop synthetic automation scores for the US market.

Most task-based studies are focussed on the US, due to the very detailed nature of the O*NET database, as well as its public accessibility. This typically forces researchers interested in the global picture or other countries and regions to base their analysis on the strong assumption of similarity of local economic and labour effects to those projected in the US context. In reality, occupational tasks can vary significantly among individual countries, which poses a particular challenge in developing countries. As such, working with tasks in the ISCO-08 system offers a useful and simple common denominator for projections across different countries, regions and income groups. Our method combines the strengths of this global framework with a significantly more detailed national classification system in Poland, which provides a detailed list of occupations and tasks, based on the latest labour market data. This helps us establish a starting point for modelling interaction of GenAI’s abilities with up-to-date tasks and occupations in a setting located between the countries with highest incomes and the emerging economies (Figure 1). Using a combination of GenAI tools, we subsequently transfer this knowledge back to the tasks in ISCO-08 system.

► 2 Tasks and occupations in the 6-digit system in Poland

Within Europe, Poland is quite representative of the larger group of Central and Eastern European (CEE) countries, albeit with a significantly larger population (38 million) than most of its peers. Globally, the CEE ranks between the high-income advanced economies and the emerging economies of the Global South. With respect to access to the internet, Poland ranks below UK and Germany, but higher than Japan and France. Thus, for assessing the potential of task automation, it seems reasonable to assume that Poland is representative of the upper threshold of automation potential, due to high availability of digital equipment and widespread internet access.

► **Figure 1. Income- and population-based similarities (A) and access to the internet (B) across countries**



The Ministry of Family, Labour and Social Policy (MOL) in Poland maintains a system of occupational classifications, with task and job descriptions that are regularly updated.² The system is used by Statistics Poland (GUS) and other state institutions for labor market analysis. At the most detailed, 6-digit level, this framework contains 2541 occupations and 29,753 corresponding tasks (Table 1). Since the system is publicly available online, we initially designed a scraping bot that crawled through individual occupational pages and downloaded job descriptions and corresponding tasks into a structured data frame.³ The extracted content was subsequently cross verified against a tabular occupational classification provided by the Ministry. The Polish classification system is compatible with the hierarchical structure of ISCO-08, which means that we could easily append an additional mapping of occupations from 1 to 4-digit level (Table 1).

² The occupational classification database (KZIS) was first published in 2014 and has been regularly updated, with the latest validation on 1 January 2025. The classification was introduced by the Regulation of the Minister of Labour and Social Policy on the classification of occupations and specializations for labor market purposes and its scope of application, dated 7 August 2014 (Dz.U. z 2018 r. poz. 227, z 2021 r. poz. 2285, z 2022 r. poz. 853, Dz.U. z 2024 r. poz. 1372). Additionally, the Ministry publishes a mapping to ISCO-08. See: <https://psz.praca.gov.pl/rynek-pracy/bazy-danych/klasyfikacja-zawodow-i-specjalnosci>.

³ Web scraping was primarily conducted to demonstrate the potential of acquiring non-traditional data from publicly accessible sources, in contexts where such activity is not restricted by server settings or legal regulations. The process adhered to responsible data collection practices, including appropriate frequency settings to avoid overloading servers.